Project Proposal: How Does an Uncued, Inferred Prior Enter the Decision Process? A Drift Diffusion Model Study of Mouse Perceptual Choice

Authors: Parth Dhodapkar (2024111009), Joshua Koilpillai (2024101128)
Course: Introduction to Neural and Cognitive Modeling (INCM) | **Topic:** Evidence Accumulation (Question 27)

1. Project Scope & Research Questions

When deciding between two sensory alternatives, the brain accumulates noisy sensory evidence over time until crossing an internal decision boundary. The Drift Diffusion Model (DDM; Ratcliff, 1978; Bogacz et al., 2006) captures this process, accounting for accuracy, full response time (RT) distributions, and the speed-accuracy trade-off (SAT).

Decisions are also shaped by prior expectations, which can enter the DDM via three distinct computational routes:

1. **Starting Point Bias (z):** Shifts initial accumulator position toward the expected boundary before evidence begins (pre-evidence motor readiness).
2. **Drift Bias (v_b):** Adds an offset to the evidence accumulation rate (asymmetric sensory evidence weighting).
3. **Dynamic Bias (b(t)):** Grows with elapsed decision time, creating an urgency-dependent reliance on the prior when sensory evidence is ambiguous (Hanks et al., 2011).

Recent literature suggests that the route depends on how the prior is generated:

- **Explicit, cued priors** (e.g., instructions that the target is 80% likely right) shift the **starting point** (Mulder et al., 2012; Hellmann & Forster, 2026).
- **Implicit sequential biases** from the immediate previous choice shift the **drift rate** across humans and rodents (Urai et al., 2019; Fernandez et al., 2025).

The Gap and Theoretical Dilemma

In ecological environments, statistical regularities are **neither explicitly cued nor reducible to 1-back motor persistence**. Instead, animals must infer latent statistical states from sequential experience. In the International Brain Laboratory (IBL, 2021) 2AFC visual task, mice experience blocks of 80/20 and 20/80 left/right stimuli without any explicit cues. While Findling et al. (2023) confirmed mice infer this prior and represent it brain-wide, they analyzed static choice probabilities without a mechanistic evidence-accumulation model. Strikingly, Fernandez, Fengler & Urai (2025) fit DDMs across species to study choice history in IBL mice, but **deliberately excluded all sessions with biased probability blocks**. Thus, how an inferred, uncued prior enters the decision process remains completely unaddressed.

Computational Modeling Protocol (Wilson & Collins, 2019)

Following feedback, rather than introducing a superficial second dataset, we ground our project in a **comprehensive parameter recovery and model recovery investigation** adhering to the rigorous guidelines

of **Wilson & Collins (2019)** ("*Ten simple rules for the computational modeling of behavioral data*", *eLife*). Before fitting empirical data, we establish parameter identifiability, quantify estimation error, rule out parameter cross-talk (e.g., z vs. v_b trade-offs), and determine the statistical power required across mouse trial counts.

Project Structure & Contributions

- **Part A: Baseline Reproduction & Wilson & Collins Validation:**
  1. Build a vectorized Euler–Maruyama DDM simulator in Python (`numpy`, `scipy`, `matplotlib`); reproduce psychometric and chronometric curves and the SAT.
  2. Conduct comprehensive parameter recovery across plausible parameter space (k, a, t_0, z, v_b, v_h, b_max, τ_b) at varying trial counts (N = 200, 500, 1000, 2000). Quantify recovery correlations (r), RMSE, linear slopes, and parameter error correlation matrices.
  3. Run model recovery across competing models to construct confusion and inversion matrices under BIC, confirming absence of model mimicry.
- **Part B: Core Scientific Contributions:**
  - **C1 (Primary Contribution): Route of the Inferred Prior.** Fit three competing mechanisms to IBL mice in biased blocks, controlling for 1-back choice history:
    - **M1 (Starting Point):** $x(0) = a \cdot z \cdot \text{prior}$ → bias strongest for **fast** RTs, then **declines**.
    - **M2 (Drift Bias):** $v(t) = vsensory + v_b \cdot prior + v_h \cdot y(\text{t-1})$ → bias is **flat** across RTs.
    - **M3 (Dynamic Bias):** $x_\text{eff}(t) = x(t) + \text{prior} \cdot b_\text{max} \cdot t / (t + \tau_b)$ → bias **increases** for slow RTs.

      Adjudicate between M1–M3 using quantitative model comparison (ΔBIC) alongside the diagnostic **Conditional Bias Function (CBF)** (White & Poldrack, 2014; Wilson & Collins Rule 6).
  - **C2 (Secondary Contribution): Urgency & Collapsing Bounds on the Wheel Task.** Compare fixed vs. Weibull-collapsing boundaries, controlling for across-trial drift (s_v) and starting-point (s_z) variabilities which can mimic urgency (Hawkins et al., 2015). Verify that the C1 prior mechanism is invariant to boundary dynamics.

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2. Novelty Statement

Prior Literature	What It Did	What It Left Open	Our Distinct Contribution
Fernandez et al. (2025)	DDM on IBL mice; choice history → drift	Excluded biased-block sessions	First DDM fit to IBL **biased-block protocol**
Findling et al. (2023)	Inferred prior decoded across mouse brain	Static logistic regression; no DDM	**Dynamic accumulation model** of inferred priors
Mulder et al. (2012)	Cued prior → starting point z	Used explicit cues in humans	Tests **uncued, inferred** prior in rodents
Hanks et al. (2011)	Dynamic time-growing prior bias	Macaque neurophysiology; primates only	First test of dynamic prior bias in **mice**

Prior Literature	What It Did	What It Left Open	Our Distinct Contribution
Chakravarty et al. (2024)	Collapsing bounds in rodent nose-pokes	Continuous wheel tasks untested	Tests fixed vs. collapsing bounds on **wheel task**
Wilson & Collins (2019)	10 rules for computational modeling	Often omitted in empirical DDM studies	**Comprehensive parameter/model recovery pipeline**

3. Planned Datasets & Methodologies

3.1 Datasets

1. **IBL Mouse Dataset (2021):** Open-access data via `one-api`. Head-fixed mice turn a wheel to report grating side at contrasts 0%, 6.25%, 12.5%, 25%, 50%, 100%. We select 10 trained mice from the biased protocol (alternating 20/80 and 80/20 blocks). For each trial, we extract signed contrast, block prior, previous choice, current choice, and RT = t_wheel_onset − t_stimulus_onset. Trials with RT < 0.1 s or > 2.5 s are excluded (5.0 s cutoff for C2).
2. **Synthetic Benchmarks:** Grids of synthetic sessions across parameter ranges (k, a, t_0, z, v_b, v_h, b_max, τ_b) and trial counts ($N \in \{200, 500, 1000, 2000\}$).

3.2 Computational Model Family & QMP Fitting

Accumulation follows $dx(t) = v(t)dt + \sigma\, dW(t)$ ($\sigma = 1$, $\Delta t = 1$ ms) with boundaries at $\pm a$. Drift is $v(t) = k \cdot$ sign(c)|c|^β^ + $v_h \cdot y_$(t-1) + [$v_b \cdot$ prior]_M2. M1 shifts $x(0) = a \cdot z \cdot$ prior; M3 adds dynamic bias prior $\cdot\, b_$max $\cdot\, t\,/\,(t + \tau_b)$; C2 introduces Weibull collapse $a(t) = a \cdot \exp(-(t/\lambda)$^γ^) and trial variabilities s_v, s_z.

Parameters are fit using Quantile Maximum Probability (QMP; Heathcote et al., 2002) across five RT quantiles {0.1, 0.3, 0.5, 0.7, 0.9} per contrast/choice condition, optimized via Nelder–Mead with fixed common random numbers.

3.3 Wilson & Collins (2019) Parameter & Model Recovery Pipeline

- **Parameter Recovery (Rule 4):** Fit synthetic datasets across 100 parameter vectors. Plot ground-truth vs. recovered parameters; quantify Pearson r, RMSE, and linear slopes ($\beta_1 \approx 1$). Compute the recovery error correlation matrix Corr($\Delta\theta$_i, $\Delta\theta$_j) to guarantee that z and v_b are statistically dissociable without trade-offs.
- **Model Recovery (Rule 5):** Simulate 50 datasets per mechanism (M1, M2, M3), fit all competing models to each dataset, and compute confusion/inversion matrices via BIC to verify > 85% discriminability.
- **Posterior Predictive Checks (Rule 6):** Simulate behavior from fitted parameters to confirm qualitative reproduction of empirical psychometric, chronometric, and CBF curves.

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4. Evaluation Strategies & Timeline

Evaluation Strategies

1. **Recovery Metrics:** Scatterplots of true vs. recovered parameters, recovery correlation $r > 0.8$, and confusion matrix diagonal dominance.
2. **Baseline Fits:** Empirical vs. simulated psychometric (sigmoid) and chronometric curves across contrasts.
3. **Prior Mechanism (C1):** Individual and summed ΔBIC across 10 mice; empirical CBF overlaid against M1 (falling), M2 (flat), and M3 (rising) predictions.
4. **Urgency & Invariance (C2):** ΔBIC for fixed vs. collapsing bounds (with *s_v*, *s_z* controls), tail distribution analysis at 5.0 s cutoff, and confirmation of C1 outcome consistency.

Timeline & Work Distribution (2 People, 4 Weeks)

Week	Parth Dhodapkar	Joshua Koilpillai	Key Deliverables
1	Implement vectorized simulator (M1–M3, collapsing bounds)	Download IBL dataset via ONE API; filter trials, extract RT/labels	Verified simulator; clean IBL dataset
2	Wilson & Collins Parameter Recovery pipeline (*N* = 200–2000)	QMP fitting pipeline; Model Recovery confusion matrix; baseline fits	Parameter & model recovery benchmarks
3	**C1:** Fit M1–M3 to 10 IBL mice; BIC analysis; compute CBF curves	**C2:** Fixed vs. collapsing fits (*s_v*, *s_z* controls); test C1 invariance	Mechanism adjudication (C1); boundary results (C2)
4	Posterior predictive validation; generate publication-grade figures	Draft report sections (Novelty, Recovery, Findings); code cleanup	Final report, figures, and code repository

Contingency Plan: C1 and the Wilson & Collins parameter recovery investigation are the core protected deliverables. If collapsing bounds (C2) encounter computational bottlenecks, C2 will be reported as an exploratory robustness subsection.

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5. Key References

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